MIL-I-7802A(AS)
14 January 1967
SUPERSEDING
MIL-I-7802(Aer)
4 February 1952

MILITARY SPECIFICATION

INDICATORS, MS28088, TURN AND SLIP (THREE PHASE)

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

* 1.1 Scope - This specification covers design and all performance requirements for procurement of turn and slip indicators operated on three phase alternating current.

* 1.2 Classification - The turn and slip indicators shall be of the following types:

MS28088-1 Type I, 26V, 3 ø, 400 cps, 2 min turn
MS28088-2 Type II, 115V, 3 ø, 400 cps, 2 min turn
MS28088-3 Type III, 26V, 3 ø, 400 cps, 4 min turn
MS28088-4 Type IV, 115V, 3 ø, 400 cps, 4 min turn

2. APPLICABLE DOCUMENTS

* 2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical

Military

MIL-P-116	Preservation, Methods of

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

Military (Continued)

MIL-G-174	Glass, Optical
MIL-C-5015	Connectors, Electric "AN" Type
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-T-18303	Test Procedures, Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

Military (Continued)

MS28088	Indicator, Turn and Slip (Three-Phase)
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

* 2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105X, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

* 3.1 Preproduction - This specification makes provision for preproduction testing.

* 3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. Parts and materials requirements shall conform to MIL-E-5400 except as otherwise specified herein.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

* 3.3 Design and Construction - The indicator shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein. The indicator shall be designed to prevent fogging on the inside of the cover glass under extremes of atmospheric conditions. The means provided shall be subject to approval by the Naval Air Systems Command. The sensitive elements of all type indicators shall be mounted so that they react only to turns about a vertical axis. Indicators shall be so constructed as to withstand the normal shocks, vibrations and such other conditions as are incident to service, shipping, storage, and installation, without failure. The indicator shall be suitable for front or rear mounting on the instrument panel.

3.3.1 Total Weight - The total weight of the indicator shall not exceed 1-3/4 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

3.3.2 Reliability -

* 3.3.2.1 Operational Stability - The indicator shall operate with satisfactory performance, continuously or intermittently for a period of at least 2500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

* 3.3.2.2 Operating Life - The indicator shall have total operating life of 25,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The indicator shall have 2000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections -

* 3.3.3.1 Cables and Connectors - The indicator shall provide for the use of cables and connectors in accordance with MIL-E-5400. The electric connector shall be a sealed connector in accordance with MIL-C-5015 and the type shall conform to MS28088.

* 3.3.3.2 Interconnection Cabling - The indicator shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the indicator.

* 3.3.4 Interchangeability of Reordered Indicators - For reordered indicators interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

* 3.3.5 Interference Control - The generation of radio interference by the indicator and the vulnerability of the indicator to radio interference shall be controlled within the limits of MIL-I-6181.

* 3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 3.3.7 Nameplate - Nameplate approval for indicator identification shall be in accordance with MIL-N-18307.

* 3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5° C)
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input power voltage	115 ±1.0 V AC or 27.5 ±0.5V AC

* 3.3.9 Service Conditions - The indicator shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

* 3.3.10 Primary Input Power Requirements - The indicator shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts:

(1) The Type II and IV indicators shall operate on AC power (three phase) 115V, 400 cps, category "B", 40 milliamperes.

(2) The Type I and III indicators shall operate on 26 V, 400 cps, proportional to category "B", 175 milliamperes.

* 3.3.10.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail Requirements -

* 3.5.1 Case - The case shall conform to MS28088.

* 3.5.1.1 Hermetic Sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case re-sealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by an atmosphere to which the instrument may be subjected in normal use in military aircraft.

* 3.5.1.2 Filling Medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ±0.1 atmosphere.

* 3.5.1.3 Body - The body shall be made either of non-ferrous low density metal, or of plastic material, shall be uniform in texture having a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service. The plastic material shall be of the thermosetting type composed of a suitable filler and phenol condensation binder.

* 3.5.1.4 Bezel Ring - The bezel ring shall be made of non-ferrous low density metal and shall be finished with a lusterless black material Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

* 3.5.1.5 Cover Glass - The cover glass shall be in accordance with MIL-G-174 (Optical Glass 523-586). Any flaws permitted by MIL-G-174 shall not interfere with reading of the dial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 3.5.2 Dial - The dial shall consist of a turn indicator and slip indicator and shall conform to Figure 1. All markings shall be durable to withstand the usage encountered in service. The style and proportions of the numerals and letters placed on the dial shall conform to MS33558.

3.5.3 Slip Indicator -

* 3.5.3.1 Indicator Tube - The slip indicator tube shall be clear annealed glass tubing free from any flaws that will affect the readability of the inclinometer. The wall thickness shall be not less then 3/64 inch and the inside of the tube shall be smooth and uniform in order that the ball will roll smoothly.

* 3.5.3.2 Indicator Ball - The ball shall be of white glass or other material approved by the procuring activity. The white color shall be in accordance with Color No. 37875 of FED-STD-595. The ball shall be a highly polished sphere 7/16 ±1/64 inch in diameter.

3.5.3.3 Damping Liquid - The damping liquid used in the slip indicator shall be colorless so that it will not interfere with the silhouette illumination of the ball by the black backing of the indicator tube.

3.5.4 Turn Indicator -

* 3.5.4.1 Pointer - The turn indicator pointer, the center mark, and the deflection marks shall conform to Figure 1. No portion of the pointer shall obstruct the view of the slip indicator tube. The pointer shall be firmly attached to the mechanism and shall be as light as practicable, and rigid to prevent oscillation under vibration, and readily adjustable.

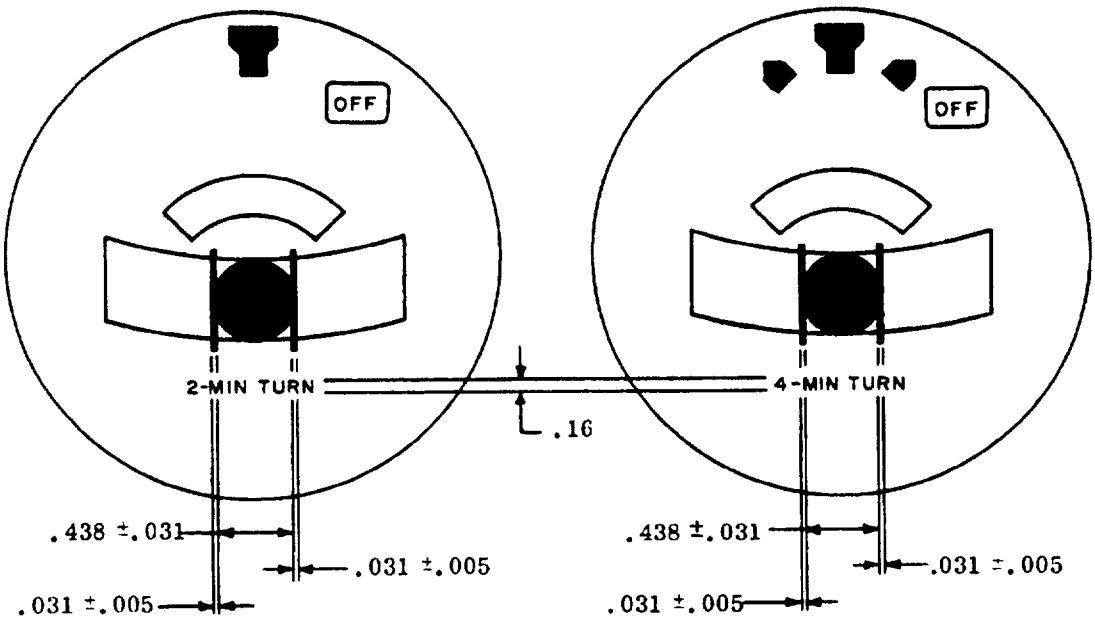
3.5.4.2 Mechanism Adjustment - Means shall be provided to adjust or correct the sensitivity and the damping of the indications. When the adjusting means is set in the neutral position, one quarter part of the total range of adjustment shall be available in either direction. This means of adjustment shall be reasonably simple to manipulate without special tools.

3.5.4.3 Gyro Parts - The gyro parts shall be mounted so that the instrument cover may be removed without disturbing the adjustments.

* 3.5.4.4 Insulation - The complete electrical system shall be insulated from the indicator case and be capable of withstanding a voltage of 500 V DC for a period of five seconds.

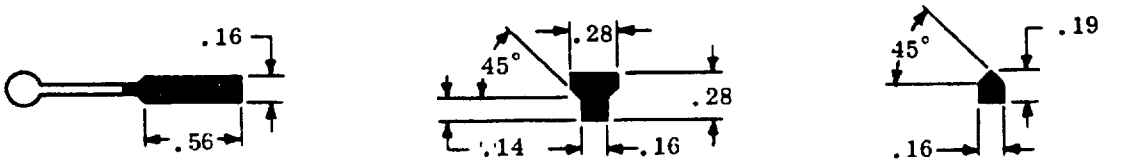
3.5.4.5 Power Failure Indicator - A suitable type indicator shall be incorporated to indicate improper phase rotation, or loss of voltage or current in any of the three phases of the gyro motor. When current or power failure occurs in any of the three phases, the indicator shall read "OFF". When power is applied and current is flowing in each leg, the flag shall disappear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.



TYPE I AND II DIAL

TYPE III AND IV DIAL



POINTER

CENTER MARK

DEFLECTION MARK

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED. TOLERANCES:
DECIMALS ±.02 ANGLES ±1/2°

MARKING	FINISH
CENTER MARK DEFLECTION MARK SHADED PORTION OF POINTER LETTERING, OFF LETTERING, 2-MIN TURN, 4-MIN TURN LINES MARKING ZERO POSITION OF BALL	LUSTERLESS WHITE COLOR NO. 37875 OF FED-STD-595
REAR HALF AND BOTTOM OF SLIP INDICATOR OR SUITABLE CARD BEHIND INCLINOMETER UNSHADED PORTION OF POINTER BACKGROUND OF DIAL POWER FAILURE FLAG	LUSTERLESS BLACK COLOR NO. 37038 OF FED-STD-595

Figure 1. Markings, dial and pointer

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Classification of Tests - Indicators covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Longevity Test

* 4.2 Preproduction Tests - Preproduction tests shall be made on an indicator representative of the production indicators to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

* 4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the indicator meets all the requirements of this specification and the contract and as a minimum shall include the tests specified herein under Test Methods (4.11).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production indicators shall be delivered prior to the approval of the preproduction sample. Prefabrication of production indicators prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of indicators to be submitted for acceptance. The preproduction sample shall not be considered as one of the indicators under the contract.

* 4.2.4 Production Indicators - Indicators supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each indicator shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for indicators already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Initial Production Tests - One of the first ten production indicators shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This indicator shall be selected by the procuring activity after the indicator has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

* 4.3.1 Scope of Tests - This indicator may be subjected to any and all tests the procuring activity deems necessary to assure that the production indicator is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements. As a minimum, the indicator shall be subjected to the following tests:

(1)	Sampling Tests	
(2)	Vibration Cycling	(4.11.26)
(3)	Radio Noise Interference	(4.11.27)
(4)	Low Temperature Exposure	(4.11.28)
(5)	High Temperature Exposure	(4.11.29)
(6)	Fogging	(4.11.30)
(7)	Acceleration	(4.11.31)
(8)	Salt Spray	(4.11.32)
(9)	Mounting Lugs	(4.11.33)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 4.3.2 Accessory Material - In addition to the complete indicator submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the indicator.

* 4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of indicators under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on indicators previously accepted when requested by the procuring activity.

* 4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the indicator shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the indicators for acceptance.

* 4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

* 4.4.1 Individual Tests - Each indicator submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each indicator accepted shall have passed the following tests:

(1)	Examination of Product	(4.11.1)
(2)	Current Consumption	(4.11.2)
(3)	Manufacturing Run-In Test	(4.11.3)
(4)	Turn Indicator Static Balance	(4.11.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

(5)	Turn Indicator Dynamic Balance	(4.11.5)
(6)	Turn Indicator Sensitivity	(4.11.6)
(7)	Turn Indicator Damping	(4.11.7)
(8)	Slip Indicator Visibility	(4.11.8)
(9)	Slip Indicator Filling	(4.11.9)
(10)	Slip Indicator Zero Position	(4.11.10)
(11)	Slip Indicator Friction	(4.11.11)
(12)	Slip Indicator Damping	(4.11.12)
(13)	Slip Indicator Sensitivity	(4.11.13)
(14)	Seal	(4.11.14)
(15)	Dielectric Strength	(4.11.15)

* 4.4.2 Sampling Tests - Indicators selected for sampling tests shall first have passed the individual tests. Indicators shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Indicators Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0 1/
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

1/ If by contract action the Initial Production Test (4.3) is deleted then a Sample Test shall be conducted on one indicator from the first 10 produced.

* 4.4.2.1 Scope of Tests - As a minimum, each indicator selected for sampling tests shall be subjected to the following tests:

(1)	Low Temperature Operation	(4.11.16)
(2)	Turn Indicator Sensitivity (Low Temperature)	(4.11.17)
(3)	Slip Indicator Damping (Low Temperature)	(4.11.18)
(4)	High Temperature Operation	(4.11.19)
(5)	Turn Indicator Sensitivity (High Temperature)	(4.11.20)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

(6) Magnetic Effect	(4.11.21)
(7) Slip Indicator Leakage	(4.11.22)
(8) Immersion (Leakage)	(4.11.23)
(9) Altitude	(4.11.24)
(10) Vibration Resonance	(4.11.25)

* 4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

* 4.4.3.1 Reliability Qualification Phase - Prior to the acceptance of indicators under the contract or order, a minimum of three (3) indicators shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of indicators to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

* 4.4.3.2 Reliability Sampling Phase Tests - Samples of the indicators shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Product on Reliability Tests". For the Sampling Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

* 4.4.3.2.1 Lot Size for Sampling Phase - The indicators constructed during one month shall be one lot.

* 4.4.3.3 Test Details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The minimum performance characteristics to be measured shall be as specified in 6.7.

* 4.4.4 Special Tests - Special tests shall be conducted on a quantity of indicators for the purpose of checking the effect of any design or material change on the performance of the indicators and to assure adequate quality control. The indicators selected for special tests may be selected from indicators previously subjected to the sampling or reliability assurance tests.

* 4.4.4.1 Special Test Schedule - Selection of indicators for special tests shall be made as follows:

(1) On an early indicator after an engineering or material change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

* 4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

* 4.4.5 Indicator Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all indicators for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Longevity Tests - Longevity Tests shall be performed in accordance with Test Plan XXVIII of MIL-STD-781. The Longevity Test procedures shall be similar to those of 4.4.3. Indicators which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for Longevity Tests.) When Reliability Tests are conducted, the Longevity Tests may be omitted if, during the Reliability Tests, two or more Indicators receive at least 2000 hours each of test time.

* 4.5.1 Test Conditions - The Longevity Test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal ground (0-5000 ft)
Humidity	Room ambient
A. C. Voltage	115 ±5 volts or (at lowest applicable frequency)
A. C. Voltage	27.5 ±2.0 volts

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the indicator temperature to stabilize. Periodically, the indicator shall be turned on and off several times and put through its various phases of operation.

* 4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

* 4.5.4 Test Data - The contractor shall keep the daily record of the performance of the indicators, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test preiod. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

* 4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

* 4.5.5 Reconditioning of Longevity Test Samples - An indicator which has been subjected to the Longevity Test shall be reconditioned as follows:

(1) On completion of the Longevity Test, the indicator shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the indicator for acceptance.

* 4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and longevity tests shall be prepared by the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

* 4.7 Reconditioning of Tested Indicators - Indicators which have been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the indicators for acceptance.

* 4.8 Presubmission Testing - No item, part or complete indicator shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

* 4.9 Rejection and Retest - Indicators which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

* 4.10 Test Conditions -

4.10.1 Standard Conditions - Unless otherwise specified, all tests required by this specification shall be made under the conditions specified in 3.3.8.

4.11 Test Methods -

4.11.1 Examination of Product - Each indicator shall be examined carefully to determine that the material and workmanship requirements have been met.

* 4.11.2 Current Consumption - A calibrated alternating current ammeter and voltmeter shall be connected between the indicator and the source of electrical power. After a period of three minutes running time, the power consumption for Type I and III indicators shall not exceed 175 milliamperes in each leg with a potential of 26 volts AC at the indicator terminals, and 40 milliamperes in each leg for Type II and IV indicators with a potential of 115 volts AC at the indicator terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

4.11.3 Manufacturing Run-In Test - Each indicator shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the indicator. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's.

The indicator shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the indicator shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the indicator in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the indicator shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future indicators. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.11.4 Turn Indicator Static Balance - The indicator, with the gyro not operating, shall be placed successively in normal operating position, 90-degree bank, and 90-degree dive, and the pointer position relative to the center mark checked. The deviation of the pointer from the zero position shall not exceed 0.015 inch for the Type I and II indicator and 0.010 inch for the Type III and IV indicator.

4.11.5 Turn Indicator Dynamic Balance - The indicator, with the gyro operating at the rated AC voltage, shall be placed successively in normal operating position, 90-degree bank, and 90-degree dive and the pointer position relative to the center mark checked. The deviation of the pointer from the zero position shall not exceed 0.015 inch for the Type I and II indicator and 0.010 inch for the Type III and IV indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

4.11.6 Turn Indicator Sensitivity - The rated power shall be applied to the gyro for a period of not less than 3 minutes. The indicator shall then be rotated about a vertical axis at the rates of turn given in Table I. The test shall be performed in both directions of rotation and the deflection of the pointer tip noted to determine that it is in accordance with the values specified in Table I.

TABLE I

TURN INDICATOR SENSITIVITY

RATE OF TURN DEGREES PER MINUTE	DEFLECTION OF POINTER TIP (INCHES)	
	TYPE I AND II	TYPE III AND IV
0	0 ± 1/64	----
36	----	1/16 ±1/64
90	5/64 ±1/32	5/32 ±1/32
180	5/32 ±1/32	5/16 ±1/32
360	5/16 ±1/16	5/8 ±1/16
1080	11/16 ±1/8	----

4.11.7 Turn Indicator Damping - The indicator shall be placed in the normal operating position on a rate table and operated for 5 minutes. The indicator shall then be rotated about its vertical axis at a rate which causes full scale deflection of the pointer and then the rotation shall be stopped suddenly. The motion of the pointer shall be so damped that the pointer returns to the zero mark without crossing it in not less than one second, and not more than 3 seconds, and without oscillation or noise other than that encountered in normal operation.

4.11.8 Slip Indicator Visibility - The indicator shall be so tipped that the ball is at rest at either end of the tube. No less than one-half of the ball shall be visible when the ball is viewed from a position 12 inches directly in front of the zero mark of the bank indicator.

4.11.9 Slip Indicator Filling - The indicator shall be so tipped that all the air in the tube is trapped in the expansion chamber end of the tube. The position of the mounting lugs of the indicator shall be in a vertical plane with the line joining the centers of the two lower mounting holes in a horizontal plane. No part of the air bubbles shall be visible when the indicator is viewed from a position 12 inches directly in front of the zero mark of the bank indicator.

4.11.10 Slip Indicator Zero Position - The indicator shall be placed in the normal operating position and the ball shall rest at the zero mark within 1/32 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 4.11.11 Slip Indicator Friction - From the level position the indicator shall be rotated about its longitudinal axis, alternately to the right and left at a rate of 2 ±1/4 degrees per second. The ball shall roll smoothly to the limits of its range and shall not stick at either end of the tube. The indicator shall not be tapped or vibrated during the test.

* 4.11.12 Slip Indicator Damping - The plane of the mounting lugs of the indicator shall be placed in a vertical position with the line joining the centers of the two lower or two upper mounting holes inclined to the horizontal at an angle of 12 degrees. The indicator shall be tipped suddenly through a vertical angle of 24 degrees, in order that the ball will roll to the opposite end of the tube. The time of roll of the ball from the zero mark of the indicator to the end of the tube shall be not less than 0.2 second.

* 4.11.13 Slip Indicator Sensitivity - From the level position the indicator shall be slowly rotated about its longitudinal axis until the ball reaches the end of the tube or to the limits of its range. At this position the angle of tilt of the indicator shall be measured and shall be within 8 to 12 degrees. This test shall be performed in both directions of rotation.

* 4.11.14 Seal - The indicator shall be tested for leaks by means of a mass spectrometer type of helium leak detector and the differential pressure existing when the test is conducted shall be the same as that to which the instrument is filled, approximately one atmosphere. The leak rate shall be based on a one atmosphere differential pressure and pure helium as the filling medium. Under these conditions, (one atmosphere differential pressure and pure helium as the filling medium), the leak rate shall not exceed 0.10 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures and helium-nitrogen mixtures, or both, shall be calculated by the manufacturer. The leak test shall be conducted either with the instrument evacuated and surrounded by helium, or after instrument has been filled. As an alternate test an immersion leak test may be performed in accordance with the requirements of MIL-E-5272, Procedure I.

* 4.11.15 Dielectric Strength - A potential of 500 volts, rms, 60 cps shall be applied between isolated pins and between pins and the case for a period of one minute. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

4.11.16 Low Temperature Operation - (This test may be combined with the Low Temperature Exposure Test, 4.11.28, at the discretion of the testing agency). The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of -54° C (-65° F) for a period of 4 hours. At the end of the 4-hour period and with the temperature maintained at -54° C (-65° F) the indicator shall be subjected to and shall meet the requirements specified in the Turn Indicator Sensitivity

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

(Low temperature) Test, 4.11.17, and the Slip Indicator Damping (Low temperature) Test, 4.11.18. There shall be no slip indicator leakage as a result of these tests.

4.11.17 Turn Indicator Sensitivity (Low Temperature) - The indicator shall be mounted in the normal operating position on a rate table and operated for 10 minutes maximum. The indicator shall then be rotated about the vertical axis at 360 degrees per minute. The pointer deflection shall be 5/16 ±1/16 inch for the Type I and II indicator, and 5/8 ±3/32 inch for the Type III and IV indicator. The rotation shall then be stopped and the pointer shall return smoothly without oscillation to the zero position within 0 ±1/64 of an inch. The test shall be performed in both directions of rotation.

4.11.18 Slip Indicator Damping (Low Temperature) - The Slip Indicator Damping Test, 4.11.12, shall be repeated and the time for the ball to travel from the zero mark to the extreme end of the tube shall be not greater than 4 seconds.

* 4.11.19 High Temperature Operation - The indicator shall be tested in accordance with Step 6, Temperature and Altitude Tests of MIL-T-5422. At the end of the 4-hour period, and with the temperature maintained at 71° C, the indicator shall be subjected to and shall meet the requirements specified in the Turn Indicator Sensitivity (High Temperature) Test, 4.11.20, and the Slip Indicator Leakage Test, 4.11.22. There shall be no slip indicator leakage as a result of these tests.

* 4.11.20 Turn Indicator Sensitivity (High Temperature) - The indicator shall be mounted in the normal operating position on a rate table and operated for 10 minutes maximum. The indicator shall then be rotated about the vertical axis at 360 degrees per minute. The pointer deflection shall be 5/16 ±1/16 inch for the Type I and II indicator and 5/8 ±1/16 inch for the Type III and IV indicator. The rotation shall then be stopped and the pointer shall return smoothly without oscillation to the zero position within 1/32 of an inch. The test shall be performed in both directions of rotation.

* 4.11.21 Magnetic Effect - The indicator shall be tested with no power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed five degrees. This test shall be repeated with power applied to the indicator. The same tolerance shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

4.11.22 Slip Indicator Leakage - The size of the bubble in the slip indicator tube shall be observed before the high temperature exposure. After the high temperature tests, the indicator shall be returned to room temperature. After a period of not more than 2 hours, the size of the bubble shall be observed to determine that it has not changed appreciably.

* 4.11.23 Immersion (Leakage) - The indicator shall be tested for leakage in accordance with Procedure I of MIL-E-5272. If the indicator incorporates attachments which are not sealed, these attachments may be removed prior to this test.

* 4.11.24 Altitude - The indicator shall be tested in accordance with Step 3, Temperature and Altitude Tests, of MIL-T-5422. When the indicator temperature has stabilized, the indicator shall be subjected to and shall meet the requirements of the Turn Indicator Sensitivity (Low Temperature) Test, 4.11.17.

* 4.11.25 Vibration Resonance - The indicator shall be tested in accordance with Part I of MIL-T-5422. Throughout the frequency ranges of this test the maximum magnitude of pointer vibration shall not exceed 1/32 of an inch, and the slip indicator ball shall not differ from its zero position by more than 1/16 of an inch.

* 4.11.26 Vibration Cycling - The indicator shall be tested in accordance with Part I of MIL-T-5422. At the completion of the test the indicator shall be subjected to and shall meet the requirements specified in the Individual Tests.

4.11.27 Radio Noise Interference - The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181.

* 4.11.28 Low Temperature Exposure - The indicator shall be tested in accordance with Step 1, Temperature and Altitude Tests, of MIL-T-5422. At the end of the exposure period the indicator shall be subjected to and shall meet the requirements of the Turn Indicator Sensitivity (Low Temperature) Tests, 4.11.17, and Slip Indicator Damping (Low Temperature) Tests, 4.11.18. The indicator shall then be returned to room temperature and allowed to saturate for a minimum period of 4 hours. It shall then be subjected to and shall meet the requirements specified in the Individual Tests.

* 4.11.29 High Temperature Exposure - The indicator shall be tested in accordance with Step 7, Temperature and Altitude Tests, of MIL-T-5422. At the end of the exposure period the indicator shall be subjected to and shall meet the requirements of the Turn Indicator Sensitivity (High Temperature) Test, 4.11.20. The indicator shall then be returned to room temperature and allowed to saturate for a minimum period of 4 hours. It shall then be subjected to and shall meet the requirements specified in the Individual Tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 4.11.30 Fogging - The indicator shall be energized at 71 ±2° C controlled ambient temperature for a minimum period of 1 hour. After this period, and while still at test temperature, an ice cube shall be rubbed on the indicator glass face for a period of from 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of internal fogging shall be cause for rejection.

* 4.11.31 Acceleration - The indicator not operating shall be mounted and accelerated, first in its vertical axis, and then in each of 2 axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g in each of the above axes for a period of 30 seconds. At the end of this acceleration, the indicator shall be subjected to and shall meet the requirements specified in the Turn Indicator Sensitivity Test, 4.11.6. No damage to the indicator shall result from the test.

* 4.11.32 Salt Spray - The indicator, with the gyro not operating but with the external connections made to simulate installed conditions, shall be tested in accordance with MIL-T-5422. At the conclusion of the test, the indicator shall be subjected to and shall meet the requirements specified in the following tests:

Current Consumption	(4.11.2)
Turn Indicator Static Balance	(4.11.4)
Turn Indicator Dynamic Balance	(4.11.5)
Turn Indicator Sensitivity	(4.11.6)
Turn Indicator Damping	(4.11.7)

4.11.33 Mounting Lugs - The indicator case shall be mounted face downward on the removable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture, and there shall be no damage to any part of the indicator.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging - The indicator shall be packaged in accordance with MIL-STD-794, Level A or C as specified in contract or order. For Level A packaging the method of preservation shall be in accordance with MIL-P-116, Method 1A-5, without preservative compound, using metal reusable containers. (See 6.2.)

* 5.2 Packing - The indicator shall be packed in accordance with MIL-STD-794, Level A, B or C as specified in contract or order. (See 6.2.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-P-7936.

5.3.1 Precautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The three-phase turn and slip indicator covered by this specification is intended for use in aircraft to indicate the rates of rotation of the airplane about its vertical axis and the lateral attitude of the aircraft relative to the apparent vertical.

* 6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) The quantity, and MS part number of the instrument desired.
(3) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

* 6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this indicator. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired that their use be investigated. When investigation shows that advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

* 6.5 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

* 6.7 Performance Characteristics To Be Measured - At least once each week the indicator shall be subjected to and shall meet the requirements of the following tests:

Turn Indicator Static Balance	(4.11.4)
Turn Indicator Dynamic Balance	(4.11.5)
Turn Indicator Sensitivity	(4.11.6)
Slip Indicator Zero Position	(4.11.10)
Slip Indicator Friction	(4.11.11)
Slip Indicator Sensitivity	(4.11.13)

A daily observation (on each workday) shall be made of each indicator in order to detect obvious failures.

* 6.8 Definitions -

* 6.8.1 Hermetic Seal - A hermetic seal shall be a perfectly

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-7802A(AS) INDICATORS, MS28088, TURN AND SLIP (THREE PHASE)

ORGANIZATION *(of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 83 1426

C-8878

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND (AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.